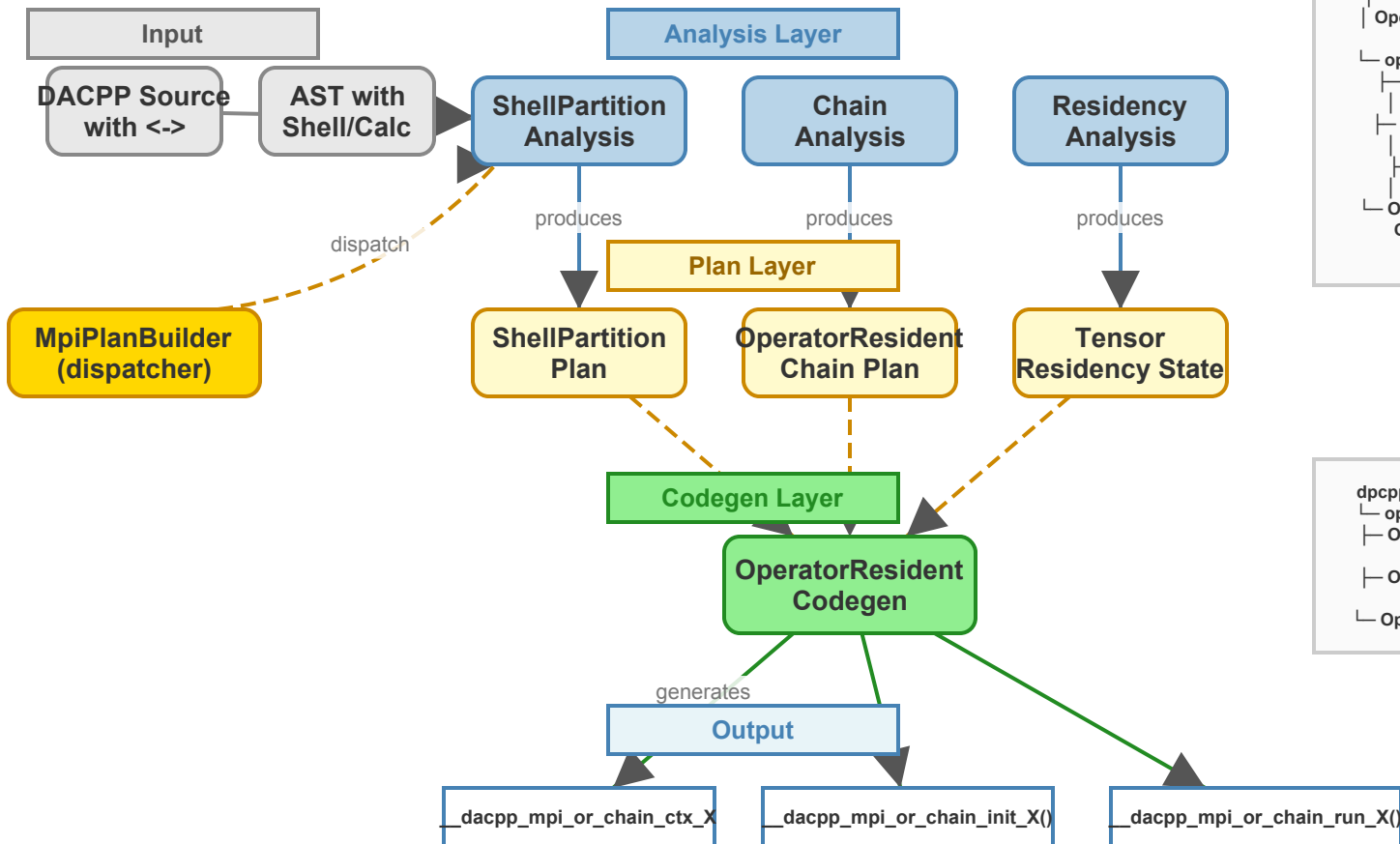


Shell-Derived Resident Partition: Implementation Architecture

Translator-side analysis and code generation for MPI operator-resident communication



```
rewriter/  
├── include/  
│   └── Rewriter_MPI_  
│       └── OperatorResident.h  
│           └── lib/mpi/  
│               └── operator_resident/  
│                   └── ShellPartition  
│                       ├── Analysis.cpp  
│                       ├── OperatorChain  
│                       │   ├── Analysis.cpp  
│                       │   ├── Residency  
│                       │   └── Analysis.cpp  
│                       └── OperatorResident  
│                           └── Codegen.cpp
```

```
dpcppLib/include/mpi/  
├── operator_resident/  
│   └── OperatorResident  
│       ├── Types.h  
│       ├── OperatorResident  
│       │   ├── Runtime.h  
│       └── OperatorResident.h
```